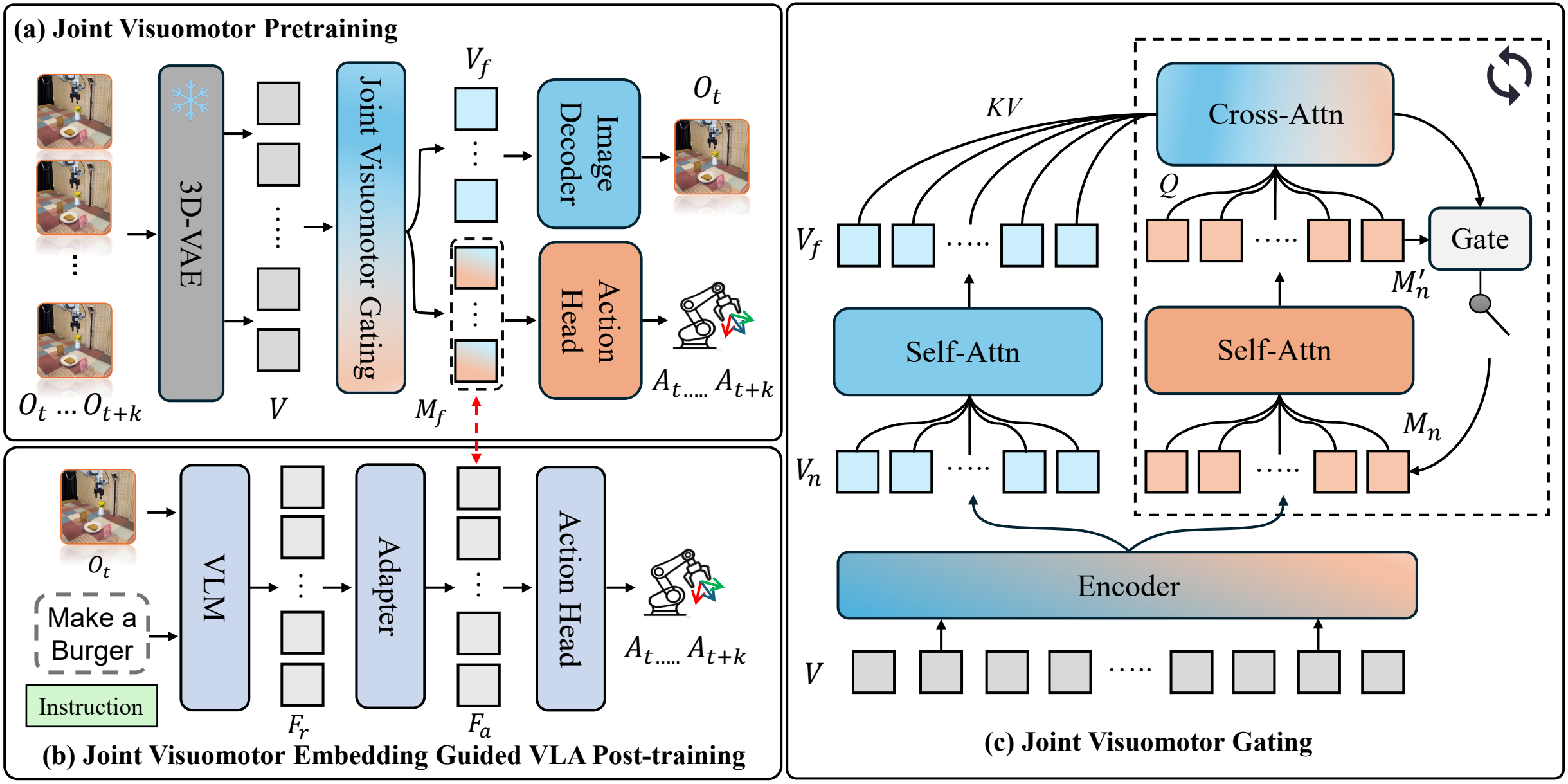
(a) Joint Visuomotor Pretraining
3D-VAE
Joint Visuomotor Gating
Image Decoder
Action Head
$O_t \dots O_{t+k}$
V
V_f
M_f
O_t
$A_t \dots A_{t+k}$
(b) Joint Visuomotor Embedding Guided VLA Post-training
O_t
Make a Burger
Instruction
VLM
Adapter
Action Head
F_r
F_a
$A_t \dots A_{t+k}$
(c) Joint Visuomotor Gating
KV
Q
Cross-Attn
Gate
V_f
V_n
M'_n
M_n
Self-Attn
Self-Attn
Encoder
V